Aleta Consultancy Services
a subsidiary of Aleta Software Labs (P) Ltd

Build Your Research Profile

"Empowering your research journey, we assist with all research-associated tasks, turning complexity into clarity."

Enhancing security in Internet of Medical Things (IoMT) Sensors with Blockchain-Enabled Framework and Differential Privacy on Edge-Fog Architectures

The Internet of Medical Things (IoMT) has transformed healthcare by enabling continuous remote monitoring, predictive diagnostics, and personalized treatment through interconnected devices such as wearables, implants, and sensors. However, this connectivity amplifies severe security risks—including data breaches, ransomware, tampering of vital readings, and unauthorized access—while strict privacy regulations (e.g., HIPAA, GDPR) demand protection of sensitive patient data (PHI/ePHI). Centralized approaches suffer from single points of failure, high latency, and privacy leakage during aggregation.

This research proposes "Enhancing Security in Internet of Medical Things (IoMT) Sensors: A Blockchain-Enabled Framework with Differential Privacy on Edge-Fog Architectures", a decentralized, proactive security solution. The framework integrates blockchain like permissioned Hyperledger Fabric or consortium models for tamper-proof data integrity, immutable audit trails, secure model aggregation, and decentralized access control/smart contracts; differential privacy to safeguard gradients and prevent inference/membership attacks during collaborative learning; and edge-fog hierarchical architectures for low-latency real-time processing.

Research Objectives

- Develop a Hybrid Blockchain-DP Architecture — Design a permissioned blockchain framework integrated with adaptive differential privacy for secure, verifiable data sharing and model updates in distributed IoMT networks, using off-chain storage for scalability.

- Incorporate Edge-Fog Deployment — Implement hierarchical processing where edge nodes handle immediate threat prevention and fog nodes manage blockchain consensus/aggregation, optimizing for low latency (<100ms), energy efficiency, and resilience against resource constraints.

- Enhance Proactive Security Mechanisms — Integrate robust defenses such as secure aggregation (e.g., Krum against poisoning), smart contract-based access control, and zero-trust principles to prevent attacks before exploitation, with formal privacy guarantees (low ε-DP values).

- Evaluate Performance and Robustness — Benchmark the framework on datasets like CICIoMT2024 and WUSTL-EHMS-2020 for metrics including detection/prevention accuracy (>98%), false positives, latency, energy consumption, privacy leakage resistance, and scalability under simulated attacks.

- Assess Practical and Ethical Implications — Analyze regulatory compliance, interoperability with existing healthcare systems, and socio-economic benefits for underserved regions (e.g., chronic monitoring in rural Karnataka), proposing deployment guidelines and ethical frameworks.